Complex Analysis Final

1. Show that if $u : \Omega \to \mathbb{R}$ is C^2 and subharmonic, and $\phi : \mathbb{R} \to \mathbb{R}$ is C^2, convex and non-decreasing, then $\phi \circ u : \Omega \to \mathbb{R}$ is subharmonic. Moreover, if u is actually harmonic, then it is not necessary to assume that ϕ is non-decreasing.

2. Use the contour integration to compute

$$\int_0^\infty \frac{1}{x^a(1+x^2)}dx, \quad -1 < a < 1.$$

3. Let f be a meromorphic function in the punctured unit disk $\mathbb{D}^*$. Suppose that there is a sequence of poles $\{z_j\}_{j=1}^\infty$ of f which accumulates to the origin. Prove that $f(\mathbb{D}^*)$ is dense in the complex plane $\mathbb{C}$.
Hint. Use the same method for proving the *Casaroti-Weierstrass Theorem*.

4. 1) Let f be a holomorphic function on the unit disk $\mathbb{D}$. Suppose that $|f(z)| \le M$ on $\mathbb{D}$ and $f(a_j) = 0$ for some $a_j \in \mathbb{D}$, $1 \le j \le n$. Show that

$$|f(z)| \le M \left|\frac{z-a_1}{1-\bar{a}_1 z}\right| \cdots \left|\frac{z-a_n}{1-\bar{a}_n z}\right|, \quad z \in \mathbb{D}.$$

Hint. Use the same method for proving the *Schwarz Lemma*.

2) A *finite Blaschke product* is a rational function of the form

$$B(z) = e^{i\varphi}\left(\frac{z-a_1}{1-\bar{a}_1 z}\right)\cdots\left(\frac{z-a_n}{1-\bar{a}_n z}\right),$$

where $a_1, \cdots, a_n \in \mathbb{D}$ and $0 \le \varphi < 2\pi$. Suppose that f is holomorphic on $\mathbb{D}$ and continuous on $\overline{\mathbb{D}}$ with $|f(z)| = 1$ for $|z| = 1$. Show that f is a finite Blaschke product.

5. Let Ω be a bounded simply-connected domain and ϕ a Riemann mapping from Ω to the unit disk $\mathbb{D}$. Set $f = \phi^{-1}$ and write $f(z) = \sum_{n=0}^\infty a_n z^n$.

1) Prove that $\text{Area}(\Omega) = \pi \sum_{n=1}^\infty n|a_n|^2$.
Hint. Integrate over $\mathbb{D}$ using polar coordinates.

2) Suppose that f extends to be analytic on $\overline{\mathbb{D}}$, $|f(z)| \ge 1$ for all $|z| = 1$, and $f(0) = 0$. Prove that $\text{Area}(\Omega) \ge \pi$.
Hint. Consider $\int_0^{2\pi} |f(e^{i\theta})|^2 d\theta$.